

The `caption` package

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Abstract

The `caption` package provides many ways to customise the captions in floating environments such `figure` and `table`. The `\rotcaption` command and the `sidewaysfigure` and `sidewaystable` environments provided by the `rotating` package from S. Rahtz and L. Barroca are also supported. The `caption` package also cooperates with the `float` package written by A. Lingnau and the `subfigure` package written by S.D. Cochran.

1 The user interface

To use this package just type

```
\usepackage[options]{caption}
```

in the preamble of your document, where the following options are supported:

<code>normal</code>	provides ‘normal’ captions, this is the default
<code>hang</code> or <code>isu</code>	provides captions with hanging indentation
<code>center</code>	provides captions where each line is centered
<code>centerlast</code>	provides captions where the last line of the paragraph is centered
<code>scriptsize</code> , ..., <code>Large</code>	sets the font size of the captions
<code>up</code> , <code>it</code> , <code>sl</code> , <code>sc</code> , <code>md</code> , <code>bf</code> , <code>rm</code> , <code>sf</code> or <code>tt</code>	sets the font attributes of the caption labels
<code>ruled</code>	supports ruled floats of the <code>float</code> package, see section 1.1 for details

`\captionsize` To setup the font size and attributes of the captions, this package also provides
`\captionlabelfont` the commands `\captionsize` and `\captionlabelfont` to allow a more flexible way to customize the captions than the above options could do. `\captionsize` is called before each caption, `\captionlabelfont` is called just before the label of the caption. Note that these commands are used by the options, e.g. the options `small` and `sf` are identical to

```
\renewcommand{\captionsize}{\small} and
\renewcommand{\captionlabelfont}{\sffamily}.
```

`\captionmargin` Furthermore there is a new length `\captionmargin` to setup an extra left and right margin for the captions, e.g. the command

```
\setlength{\captionmargin}{10pt}
```

sets this margin to 10pt.

1.1 The rotating, float and subfigure package

If you want to use this package together with the `rotating`[1], `float`[2] and/or the `subfigure` package, you have to input this package *after* the other ones, like

```
\usepackage{float,rotating,subfigure}
\usepackage[centerlast,small,sc]{caption}
```

The `caption` package now redefines the `sidewaysfigure` and `sidewaystable` environments and the `\rotcaption` command provided by the `rotating` package from S. Rahtz and L. Barroca. Note that the `\captionmargin` does not affect the `\rotcaption` command.

It also redefines the captions of the `plain` and `boxed` styled floats provided by the `float` package from A. Lingnau. `ruled` floats are not supported by default, but you can change this via setting the option `ruled`. Note that the `\captionmargin` is not supported in ruled floats.

If the `subfigure` package from S.D. Cochran is detected, the options `scriptsize`, ..., `large` will redefine `\subcaptionsize` in an adequate way. If you redefine `\captionsize` by yourself and use the `subfigure` package, you also have to redefine `\subcaptionsize` by yourself. Also the commands `\@thesubfigure` and `\@thesubtable` will be redefined to use the `\captionlabelfont` command, please take this into consideration if you redefine `\@thesubfigure` or `\@thesubtable` by yourself. E.g. a adequate version of the second example in `subfigure.sty` will be:

```
\renewcommand{\thesubfigure}{\thefigure.\arabic{subfigure}}
\makeatletter
\renewcommand{\@thesubfigure}{\captionlabelfont\thesubfigure:}\space}
\renewcommand{\p@subfigure}{}
\makeatother
...
```

This package cooperates with the versions 2.0 (1994/04/15) and 2.5 (1994/11/11) of the `rotating` package, version 1.2 (1994/06/20) of the `float` package and version 1.6 (1993/05/13) of the `subfigure` package, but will hopefully work with future versions, too.

2 Example

Here's an example figure which was produced with the following code in the preamble of this document:

```
\usepackage[hang,small,sf]{caption}
\setlength{\captionmargin}{20pt}
```

EXAMPLE FIGURE

Figure 1: This is an example caption with a small font and a sans serif label. The hang option was used. There is a left and right margin of 20pt.

3 The code

3.1 Identification

```
1 \NeedsTeXFormat{LaTeX2e}[1994/06/01]
2 \ProvidesPackage{caption}[1995/01/09 v1.3 caption package (AS)]
3 \typeout{Package: caption v1.3 <1995/01/09> (Axel Sommerfeldt)}
```

3.2 Initial code

```
\captionsize
\captionlabelfont
\captionmargin
4 \newcommand{\captionsize}{}
5 \newcommand{\captionlabelfont}{}
6 \newlength{\captionmargin}
```

Here are the different basic types of captions implemented:

```
\as@normalcaption The 'normal' caption
7 \newcommand{\as@normalcaption}[2]{%
8   #1 #2}

\as@isucaption The 'iso' or 'hang' caption; this code was taken from The LATEX Companion[3,
p155] and modified
9 \newcommand{\as@isucaption}[2]{%
10   \sbox{\as@captionbox}{#1\ }%
11   \addtolength{\as@captionwidth}{-\wd\as@captionbox}%
12   \usebox{\as@captionbox}\parbox[t]{\as@captionwidth}{#2}}

\as@centercaption The 'center' caption
13 \newcommand{\as@centercaption}[2]{%
14   \parbox[t]{\as@captionwidth}{\begin{center}#1 #2\end{center}}}
```

`\as@annecaption` The ‘centerlast’ caption; the idea how to do this was taken from Brüggemann-Klein[4], it is also mentioned in Kopka[5, p227]

```
15 \newcommand{\as@annecaption}[2]{%
16   \addtolength{\leftskip}{0pt plus 1fil}%
17   \addtolength{\rightskip}{0pt plus -1fil}%
18   \setlength{\parfillskip}{0pt plus 2fil}%
19   #1 #2}
```

3.3 Detection of the subfigure package

`\as@subcapsize` If the subfigure package is loaded, a little message will be typeout and `\as@subcapsize`
`\@thesubfigure` – which sets the size of the subcaptions – will be defined. Also `\@thesubfigure`
`\@thesubtable` and `\@thesubtable` will be redefined here to support the `\captionlabelfont` in subcaptions, too (thanks to Kevin Ruland for this idea!). If you don’t like this behaviour, just load the `caption` package *after* the `subfigure` package (and eventually redefine the `\subcapsize` by yourself) or redefine `\@thesubfigure` and `\@thesubtable` after loading the `caption` package as shown in the documentation of the `subfigure` package.

```
20 \ifx\thesubfigure\undefined
21   \newcommand{\as@subcapsize}[1]{%
22     \else%
23     \typeout{\space\space\space\space\space\space\space\space\space\space
24       ‘subfigure’ package detected}
25     \let\as@subcapsize\subcapsize
26     \renewcommand{\@thesubfigure}{\captionlabelfont\thesubfigure}\space}
27     \renewcommand{\@thesubtable}{\captionlabelfont\thesubtable}\space}
28 \fi
```

3.4 Declaration of options

There are four different types of captions supported: `normal`, `isu`, `center` and `centerlast`. `hang` is exactly the same as `isu`.

```
29 \DeclareOption{normal}{\let\as@caption\as@normalcaption}
30 \DeclareOption{isu}{\let\as@caption\as@isucaption}
31 \DeclareOption{hang}{\ExecuteOptions{isu}}
32 \DeclareOption{center}{\let\as@caption\as@centercaption}
33 \DeclareOption{anne}{\let\as@caption\as@annecaption}
34 \DeclareOption{centerlast}{\ExecuteOptions{anne}}
```

There are options for six different font sizes available, they also redefine the `\subcapsize` provided by the `subfigure` package (if detected).

```
35 \DeclareOption{scriptsize}{%
36   \renewcommand{\captionsize}{\scriptsize}
37   \as@subcapsize{\scriptsize}}
38 \DeclareOption{footnotesize}{%
39   \renewcommand{\captionsize}{\footnotesize}
40   \as@subcapsize{\scriptsize}}
41 \DeclareOption{small}{%
```

```

42 \renewcommand{\captionsize}{\small}
43 \as@subcapsize{\footnotesize}
44 \DeclareOption{normalsize}{%
45 \renewcommand{\captionsize}{\normalsize}
46 \as@subcapsize{\footnotesize}}
47 \DeclareOption{large}{%
48 \renewcommand{\captionsize}{\large}
49 \as@subcapsize{\normalsize}}
50 \DeclareOption{Large}{%
51 \renewcommand{\captionsize}{\Large}
52 \as@subcapsize{\large}}

```

There are nine options available to set the font attributes of the caption labels.

```

53 \DeclareOption{up}{\renewcommand{\captionlabelfont}{\upshape}}
54 \DeclareOption{it}{\renewcommand{\captionlabelfont}{\itshape}}
55 \DeclareOption{sl}{\renewcommand{\captionlabelfont}{\slshape}}
56 \DeclareOption{sc}{\renewcommand{\captionlabelfont}{\scshape}}
57 \DeclareOption{md}{\renewcommand{\captionlabelfont}{\mdseries}}
58 \DeclareOption{bf}{\renewcommand{\captionlabelfont}{\bfseries}}
59 \DeclareOption{rm}{\renewcommand{\captionlabelfont}{\rmfamily}}
60 \DeclareOption{sf}{\renewcommand{\captionlabelfont}{\sffamily}}
61 \DeclareOption{tt}{\renewcommand{\captionlabelfont}{\ttfamily}}

```

If the option `ruled` is set, the captions of ruled floats provided by the `float` package will also be supported.

```

62 \DeclareOption{ruled}{\newcommand\as@ruled{}}

```

3.5 Execution of options

The ‘normal’ type of caption is preselected.

```

63 \ExecuteOptions{normal}
64 \ProcessOptions

```

3.6 Main code

```

\as@captionbox The new \@makecaption code
\as@captionwidth 65 \newsavebox{\as@captionbox}
\as@makecaption 66 \newlength{\as@captionwidth}
\@makecaption 67 \newcommand{\as@makecaption}[2]{%
68 \setlength{\leftskip}{\captionmargin}%
69 \setlength{\rightskip}{\captionmargin}%
70 \addtolength{\as@captionwidth}{-2\captionmargin}%
71 \captionsize%
72 \sbox{\as@captionbox}{\captionlabelfont #1:} #2}%
73 \ifdim \wd\as@captionbox >\as@captionwidth
74 \as@caption{\captionlabelfont #1:}{#2}\par%
75 \else%
76 \begin{center}\usebox{\as@captionbox}\end{center}%
77 \fi}

```

```

78 \renewcommand{\@makecaption}[2]{%
79   \vskip\abovecaptionskip%
80   \setlength{\as@captionwidth}{\linewidth}%
81   \as@makecaption{#1}{#2}%
82   \vskip\belowcaptionskip}

```

3.7 Support of the rotating package

`\@makercaption` If the rotating package is loaded, the commands `\@makerotcaption` (for support of `\rotcaption`) and `\@makercaption`, furthermore the environments `sidewaysfigure` and `sidewaystable` will be redefined here. The code was taken from the rotating package [1] itself and adapted.

```

83 \ifx\@makercaption\undefined
84 \else
85   \typeout{\space\space\space\space\space\space\space\space\space\space
86     'rotating' package detected}
87   \renewcommand{\@makecaption}[2]{%
88     \vskip 10\p%
89     \setlength{\as@captionwidth}{\textheight}%
90     \as@makecaption{#1}{#2}}
91   \renewcommand{\@makerotcaption}[2]{%
92     \captionsize%
93     \sbox{\as@captionbox}{\captionlabelfont #1:} #2}%
94     \ifdim \wd\as@captionbox > .8\vsiz
95       \rotatebox{90}{%
96         \setlength{\as@captionwidth}{.8\textheight}%
97         \begin{minipage}{\as@captionwidth}%
98           \as@caption{\captionlabelfont #1:}{#2}%
99         \end{minipage}}\par
100     \else%
101       \rotatebox{90}{\usebox{\as@captionbox}}%
102     \fi
103     \hspace{12pt}}
104   \renewcommand{\sidewaysfigure}{%
105     \let\@makecaption\@makercaption\@rotfloat{figure}}
106   \renewcommand{\sidewaystable}{%
107     \let\@makecaption\@makercaption\@rotfloat{table}}
108 \fi

```

3.8 Support of the float package

`\floatc@plain` If the float package is loaded, the command `\floatc@plain` (and the command `\floatc@ruled` if the option `ruled` is given) will be redefined here.

```

109 \ifx\floatc@plain\undefined
110 \else
111   \typeout{\space\space\space\space\space\space\space\space\space\space
112     'float' package detected}
113   \renewcommand\floatc@plain[2]{%

```

```

114   \setlength{\as@captionwidth}{\linewidth}%
115   \as@makecaption{#1}{#2}}
116 \ifx\as@ruled\undefined
117 \else
118   \renewcommand\floatc@ruled[2]{%
119     \setlength{\as@captionwidth}{\linewidth}%
120     \captionsize%
121     \as@caption{\captionlabelfont #1:}{#2}\par}
122 \fi
123 \fi

```

References

- [1] Sebastian Rahtz and Leonor Barroca: *A style option for rotated objects in L^AT_EX*, 1994/08/26
- [2] Anselm Lingnau: *An Improved Environment for Floats*, 1994/06/20
- [3] Michel Goossens, Frank Mittelbach and Alexander Samarin: *The L^AT_EX Companion*, Addison-Wesley, Reading, Massachusetts, 1994.
- [4] Anne Brüggemann-Klein: *Einführung in die Dokumentverarbeitung*, B.G. Teubner, Stuttgart, 1989
- [5] Helmut Kopka: *L^AT_EX – Erweiterungsmöglichkeiten*, 3. überarbeitete Auflage, Addison-Wesley, Bonn, 1991